

Raaghav Thirumaligai

Prototyping Designer ~ Mechanical Engineer ~ Controls PhD Student

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PROFILE

Prototyping designer and mechanical engineer exploring how new hardware can create new interactions. Builds physical mock-ups, motion studies, and functional prototypes by moving between sketching, CAD, fabrication, sensing, and code. Product-design training and controls research bring a systems view of feedback, behavior, and how people and devices respond to one another.

CAPABILITIES

Interaction Prototyping – Physical mockups, haptic and touch concepts, sensors, input/output behaviors
Visual Communication – Hand sketches, Apple Keynote storyboards and animation, CAD renders
Software & Interactive Systems – Python, OpenCV, Arduino, MATLAB/Simulink, sensors, DAQ
Product Development – SolidWorks, Fusion 360, 3D printing, sheet metal, drawings, GD&T, FEA
Systems & Experiments – Feedback, system identification, estimation, test design, design-build-test

SELECTED PROJECTS

Rapid Interaction Studies: Disc, Orb & Hover Touch

Aug 2026

- Developed 3 speculative hardware-interaction concepts through hand sketches, Keynote storyboards, animation, and a rapid physical mockup.
- Built a same-day cardboard Disc prototype to test palm scale, grip, press location, thickness, and display glare before committing to CAD or code.

Autonomous Can-Collection Robot (C.L.E.A.N.): Team Lead

Apr 2024 - Jun 2024

- Led a 5-person team through weekly design-build-test cycles, culminating in an autonomous robot that could find, approach, identify, grasp, and lift aluminum cans.
- Trained and tuned an OpenCV cascade classifier on approximately 1,500 custom images; converted the detected can's bounding-box centroid into steering commands on a Raspberry Pi.
- Integrated a custom 3D-printed treaded chassis and 2-DOF worm-gear arm/claw with motor drivers, an Arduino, and ultrasonic and inductive sensors.

Collegiate Electric Race Car (Formula SAE EV): Vehicle Systems Lead

Aug 2021 - Jun 2025

- 4-year contributor and Internal VP/VP on an electric race-car design-build-test team; led work across vehicle systems and coordinated teammates through build cycles.
- Designed and built a 600 V accumulator container using an adhesive-first joining strategy; produced CAD, engineering drawings, and structural checks.
- Led braking-system design, manufacturing, and testing; separately won an undergraduate research grant to characterize brake-fluid viscosity and presented the results.

EDUCATION

- **University of California, Santa Barbara - Ph.D. Mechanical Engineering** **Projected Jun 2029**
GPA 3.91/4.0 — Control Theory — Year 1 completed; available to take academic leave for full-time work
- **University of California, Santa Barbara - M.S. Mechanical Engineering** **Jul 2025**
GPA 3.90/4.0 — Project: [Clogging of fiber suspensions at constrictions](#)
- **University of California, Santa Barbara - B.S. Mechanical Engineering** **Jun 2025**
GPA 3.76/4.0 — Product Design — Capstone: [600 V Vehicle Battery Container](#)

RESEARCH EXPERIENCE

Biological Control Lab, Yeung UCSB - *Research Assistant*

Jan 2025 - Jan 2026

- Co-developed a control-oriented model of genome-wide transcription coupled to DNA supercoiling dynamics regulated by topoisomerases.
- Supported analysis of stochastic and nonlinear population models (stability/sensitivity analysis) to identify parameters influencing transcription and growth dynamics.
- Co-author on manuscript accepted (Jan 2026); preprint: [10.1101/2025.05.26.655193](https://arxiv.org/abs/10.1101/2025.05.26.655193).

M&M Flow Lab, Sauret UCSB - *Master's Project*

Aug 2024 - Jun 2025

- Researched clogging dynamics in particle-laden suspensions in constricted flows, relevant to medical and industrial systems.
- Prototyped a vision-based clog detector using a rolling brightness/windowed-visibility metric; used it to support logging and post-analysis of clogging events.
- Presented findings at November 2024 APS Division of Fluid Dynamics meeting.

M&M Flow Lab, Sauret UCSB - *Undergraduate Researcher*

Jan 2023 - Aug 2024

- Explored cohesive forces in granular materials with 2 PhD student mentors to help understand inter-particle mechanics from the introduction of cohesive forces.
- Designed and built an instrumented annular shear cell; developed an Arduino-MATLAB DAQ + calibration workflow for strain-gauge measurements and repeatable experiments.
- Presented work at Southern California Flow Physics Symposium.

Fluid Energy Science Lab, Luzzatto-Fegiz UCSB - *Undergraduate Researcher*

Jun 2023 - Oct 2023

- 3D-printed and laser-cut annuli to simulate kites for wind energy applications.
- Data collection for wake model verification in a wind tunnel; made an Arduino-MATLAB DAQ system and app.

EXPERIENCE - WORK

Teaching Assistant - UCSB: Mech E Academic Student Employee

Apr 2024 - Present

- Taught laboratory methods, dynamics, vibrations, and fluids through lab sections, office hours, and course assessment.
- Explained unfamiliar technical ideas, coached bio-inspired design teams, and gave actionable feedback across different learning styles.

Resident Assistant - UCSB: San Rafael Residence Hall

Aug 2022 - Jun 2023

- Managed resident support, conflict resolution, and event planning in a high-traffic dorm environment.

GameMaster - Diablo Escapes, Walnut Creek

Jul 2020 - Aug 2022

- Guided customers through VR escape rooms; troubleshoot hardware/software issues and explained unfamiliar tech.
- Helped design puzzles for physical escape rooms under construction.

HONORS

- Undergraduate Engineering Honors Program
- Dean's Honors (Engineering), 5 quarters
- 2 Undergraduate Research Awards (\$3750 and \$2500)

Sept 2023 - Jun 2024

2024 and 2025

Last updated: August 10, 2026